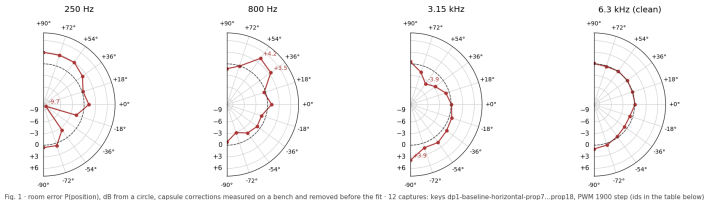


An overview of our chamber problem

Two errors, not one: the blade tones are bent by up to 8 dB below 1 kHz, and under them sits a broadband error of ± 1.3 dB (3.3 dB worst) that no band is exempt from. The arc is not qualified to a standard anywhere yet. For these runs the arc was laid flat (horizontal), with the propeller axis vertical. All microphones then sit at the same height in the propeller plane, so a perfect setup reads the same level everywhere: a circle. Twelve runs (prop7–18, 1–2 Sept 2026, PWM 1900); what is left after each microphone's own offset is removed is the room.

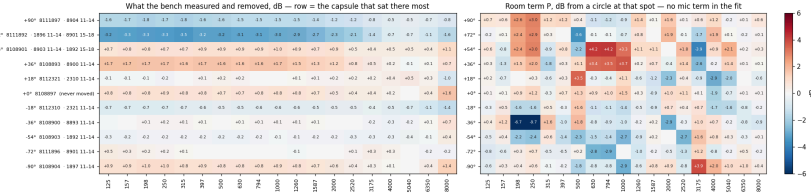
2026-09-17 · rev 3
data: laptop backup of 2 Sept
scripts: arc_error_map.py, arc_validation_diagnostics.py
full guide: docs/arc-validation-guide.html
remedies report: docs/arc-validation-remedies.pdf



What the room does (right panel of Fig. 2)

- How to read the map.** Row = spot on the arc, column = frequency band. White: the spot reads what a circle predicts. Red: too loud. Blue: too quiet. Each microphone's own error was measured on a bench and taken out before the fit (left panel), so the colour is the room.
- Deep blue at -72° and red at $+36^\circ/+54^\circ$, 500-1000 Hz.** A sound wave that bounces off a nearby surface arrives a little later than the direct wave. If the extra path is half a wavelength the two cancel and the spot goes blue; if it is a whole wavelength they add and the spot goes red. The extra path depends on the spot, so neighbours can be opposite colours and one spot can flip colour from band to band. Wavelength here is 35–70 cm, so the surface is within about a metre.
- The 200-250 Hz column is the blade tone, and it is not one nearby surface.** At BPF 216 Hz the tone polar is round to ± 2 dB; at 238 Hz -36° reads -8 dB; at 258 Hz the hole moves to -72° and the peak to $+54^\circ$, while the broadband floor beside the tone stays round to ± 1 dB. A surface within a metre cannot change the pattern that fast; narrow room modes sampled by a line spectrum can, and so can a propeller whose inflow is not axisymmetric. A loudspeaker at the hub decides which (remedies report, §03, §10).
- Blue on one side, red on the other: the surface is off-centre.** A surface symmetric around the propeller (the floor under a flat arc, say) gives every microphone the same extra path, so a whole row would be one colour. Here the colour changes around the arc, so the microphones are at different distances from the surface: it sits to one side. The colours do not say which side. Probing with the tone at three speeds: the $+36^\circ/+54^\circ$ reflection has an extra path of 1.5 m and a hard surface's amplitude, the -72° one 0.7 m (remedies report, Fig. 3); the unrecorded arc radius is what turns those into places.
- Taking the tones out removes the big errors, not all of them.** A wide band averages a reflection out, a tone does not, which is why the tones carry the big errors. With every tone notched the polar is round to ± 1.3 dB in 84 % of cells, worst 3.3 dB — but 90 of those 209 cells sit more than three standard errors from zero, so the rest is a real pattern, not scatter, and it is no smaller above 3 kHz than below (87 % of cells inside ± 1.3 dB below 3 kHz, 76 % above) (3.3 dB at $+72^\circ/2$ kHz, 3.2 at $+18^\circ/2$ kHz, 3.1 at $+54^\circ/5$ kHz). The tones need the loudspeaker check or the speed ladder; the broadband floor of ± 1 –3 dB is there in every band (report §00, §03).

Fig. 2 · What the bench measured and removed (left) and what the room leaves behind (right), same scale



What to look for in the chamber

Three suspects, one per band in Fig. 2. Each is a size and a place, read off the map: a surface shows up once the wavelength is shorter than it (Rajmane and Baumann, DAGA 2016, Table 1: a 35 cm plate from 1 kHz, a 10 cm plate from 3.15 kHz).

- 200-250 Hz, blue at -36° .** Something about 1.5 m across, off to one side of the rig, with a bounced path 0.7 m longer than the direct one at -36° : a wall, a table top or the corner within about half a metre of that microphone.
- 630-1000 Hz, blue at -72° .** Something about 0.4 m across near the -90° end of the arc on the corner side: the stand base, a bar of the arc frame, a cable tray.
- 3150 Hz, scattered red and blue.** Something about 0.1 m across at the microphones themselves: clamps, mounts, the cable loop.

How the numbers were made (not a sum, not an average)

Every level in the twelve runs (11 microphones $\times$ 19 third-octave bands $\times$ 12 runs) is written as two parts added together: **run gain + position error**. Both are found by least squares over all 12×11 levels in every band. **Run gain** takes up any level difference between runs (RPM, day, load). **There is no microphone term**: each capsule's own error was measured on a bench against the other ten and folded into its calibration file, so it is already out of the levels before the fit sees them. **Position error** is what is left that belongs to a spot on the arc; it is forced to average zero over the eleven spots, so it reads as dB from a circle. That is what Figs. 1 and 2 show.

The turned-around runs are handled before the fit: for prop11–14 the label +e is entered as physical $-e$, so " -72° " always means the same spot in the room. Until 16 Sept 2026 the microphone offset was estimated alongside the position error, and the two can be told apart only where one spot was covered by different microphones. Ours covered five disconnected clusters, so four directions of the model were undetermined and every comparison between clusters carried an unmeasured offset of about 1 dB. Measuring the capsules on a bench removes that: the map below is determined, up to one overall offset that no method here can fix. Spectra: Welch 4096-point Hann, UMIK-2 calibration as the app applies it, third-octave sums 125 Hz–8 kHz. Data: the last PWM-1900 capture in each key 2004_6in_unset_dp1-baseline-horizontal-prop7..prop18, 11 mics $\times$ 2.0 s (prop1–6 and the 31 Aug keys match neither orientation and are left out); ids in docs/analysis/arc-validation-data.json. Code: scripts/arc_error_map.py --no-mic-term, function fit; capsule corrections calibrator/apply_corrections.py; data pack scripts/build_arc_data_pack.py.

Do the numbers add up?

All numbers re-checked against the source data 2026-09-11, 40 of 40 (AUDIT-2026-09-11.md). Refitting reproduces the maps to 0.01 dB. 132 cells per band are explained by 23 unknowns (34 before the microphones were measured); 87% of them are within 2 dB of gain + P + M. Scatter is 0.6–0.8 dB in the quiet bands and 1.5–2.3 dB in 200–250 Hz and 500–1000 Hz, the bands where the room acts. The worst cells (+11 dB at $-36^\circ/250$ Hz; +6 to +7 dB at $0^\circ/2$ –4 kHz) come from prop7, which spun at BPF 164 Hz against 238 Hz elsewhere; leaving it out moves the maps by under 1 dB. **Resolved 2026-09-16:** a microphone separates a bad spot from a bad capsule only where it sat in both, and ours visited five disconnected clusters, so each spot against the arc mean used to carry an unmeasured decibel. Measuring the capsules against each other on a bench removed that; only one overall offset is left (report, Problems 4 and 6).

What we ruled out

- The microphones — measured now, not inferred.** On 16 Sept 2026 all eleven capsules were compared against each other on a bench: every capsule against every other at one seat, 55 pairs, agreeing to 0.08 dB on repeat. This had to be done off the arc, because only one microphone pair ever changed places there ($+72^\circ/+54^\circ$, after prop10), which left the fit unable to tell a bad microphone from a bad spot across five clusters of positions. **Seven of the eleven factory calibration files were wrong by 0.6 to 3.6 dB** — worst 811-1892 at -3.6 dB, whose capsule is in fact flat to 0.24 dB against the four files that check out; only its certificate was wrong. All eleven files were corrected (left panel of Fig. 2), and the room map is now fitted with **no microphone term at all**. Dropping eleven free parameters cost 0.14 dB of residual, which is how we know the correction holds.
- A labelling error.** prop11–14 were measured with the arc turned around. Their patterns match the mirror image of the other runs (correlation +0.79 to +0.98). Plotted by label the dip appears on the other side; plotted by physical position they agree with the other eight runs. Turning the arc moves the labels, not the blue and red patches.
- The propeller's mean field.** With the arc flat and the axis vertical every microphone sees the propeller from the same angle, so an axisymmetric propeller cannot make one spot differ from another. Its inflow can, for the tones only; at 238 Hz the tones are bent and the broadband beside them is not, so this is not excluded (remedies report, §10).

Next test (one afternoon)

- Put a small loudspeaker on the stand at the hub. Play a 20 Hz–20 kHz sine sweep and record all microphones; each microphone's own impulse response gives its echo delays (no clock sync needed), times 343 m/s, the extra paths. Then play tones at 216, 238 and 258 Hz: if the ± 8 dB pattern of Fig. 1 comes back, it is the room; if not, it is the propeller's inflow.
- Write down the geometry in docs/mic_arc.md: arc radius, propeller height, which wall or corner is at the -90° end, distances from the -72° and $+72^\circ$ microphones to the nearest wall.
- Run a ladder of 12–15 motor speeds (BPF every ~ 5 Hz, 205–265 Hz) and average the tone polar over it; cover or move the surfaces the sweep names; rerun the fit. Foam works from 650 Hz up; at 250 Hz the levers are averaging, correction and distance, then resonant absorbers.

Worked example · -72° spot at 800 Hz, two mics, two orientations

RUN	MIC THERE	MEASURED	RUN GAIN	P ROOM	MODEL	LEFT OVER
prop18	811-1896	59.1	62.4	-2.9	59.6	-0.5
prop12 (180°)	810-8901	57.7	62.3	-2.9	59.4	-1.7

Two different capsules, the arc turned between the runs, one room value. There is no microphone column any more: each capsule's own error was measured on a bench and taken out of the levels before the fit, so what the model does not explain is run-to-run scatter, not an unknown mic.